



CRTMS

CENTRAL RAILWAY TRACTION MANAGEMENT SYSTEM

One secure platform. Real-time visibility across Staff, Crew, Safety, and Analytics — built for TRO Mumbai Division



“Intelligent Supervision for Safer Traction Operations”

01

Administrative
Control

02

Digital
Detail Book

03

Safety &
Analytics

04

Crew
Counselling

01 ► ADMINISTRATIVE CONTROL & WORKFORCE VISIBILITY

Real-Time Dashboard [LIVE]

The CRTMS Dashboard delivers a live operational snapshot — total staff strength across lobbies, PME due monitoring, attendance, and leave tracking — with quick access to reports and administrative actions.

Comprehensive Staff Bio-Data [LIVE]

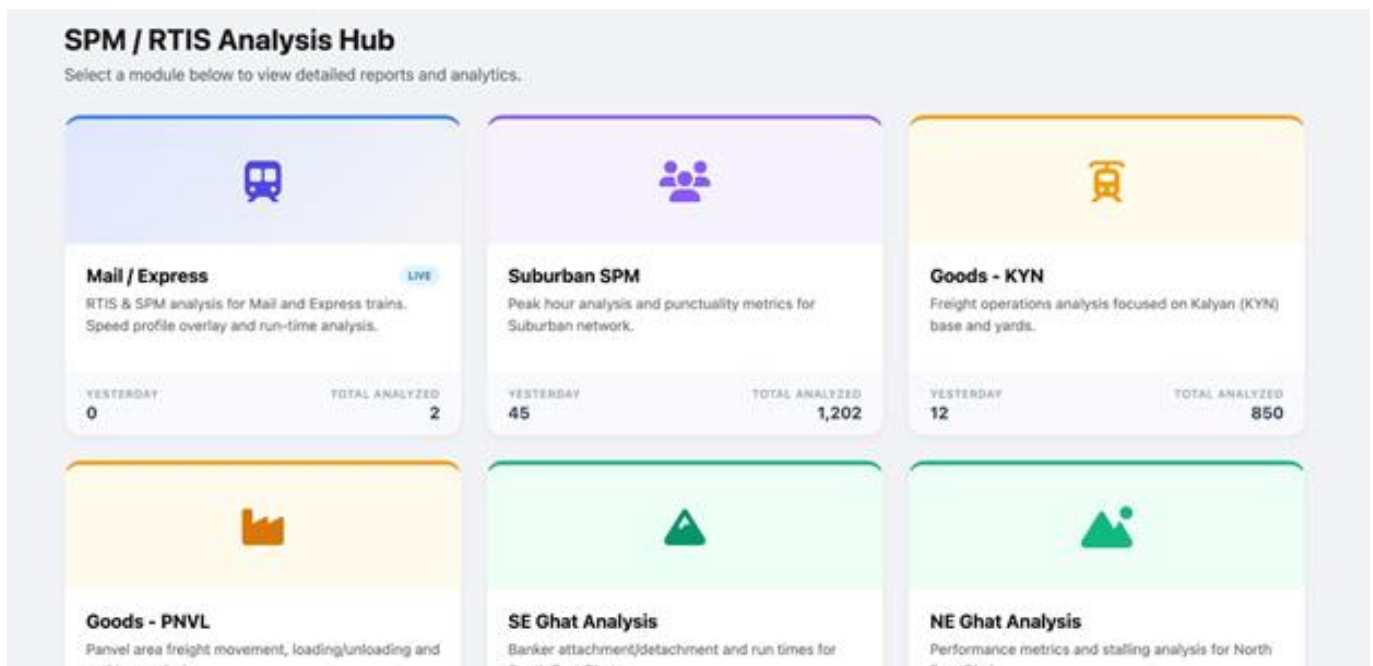
Centralized professional records for every staff member, instantly searchable by HRMS ID, CMS ID, PF Number, or name.

CLI Nomination & Safety Category Management [LIVE]

Previously maintained through manual Word files, CLI nominations and safety category changes are now fully integrated into CRTMS — ensuring real-time database updates and complete record integrity.

Midnight Position Monitoring [IN PROGRESS]

Tracks 00:00 hrs crew availability, sanctioned vs. on-roll strength, and non-effective staff with historical position records — replacing the current Excel-based approach with a structured digital platform.



02 ► DIGITAL DETAIL BOOK

Phase II • Goods Lobbies: KYN & PNVL

Daily crew booking at Goods Lobbies currently relies on physical whiteboards and manual registers. Phase II introduces a fully digital crew management system:

- Digital preparation and display of daily crew booking
- Automatic tracking of continuous night duty hours and weekly rest
- Visibility of Learning Road progress
- Signal Sighting and AWS Monitoring Reports

Improving transparency, compliance, fatigue monitoring, and ease of supervision across all Goods Lobbies.

03 ► UNIFIED SAFETY & OPERATIONAL ANALYTICS

Single Integrated Platform — SPM & RTIS [LIVE]

CRTMS consolidates Freight, Ghat, and Suburban SPM data along with RTIS inputs for Mail and Goods trains into one unified analytical platform — enabling comprehensive, data-driven driving skill monitoring.

One-Click Braking Pattern Analysis [LIVE]

Fortnight braking reports for Mail/Express and Suburban trains — previously prepared manually — are now generated in a single click. Analysis can be filtered by any station, direction, specific Loco Pilots, or critical halts such as Up Ghat sections.

Speed Violation Analytics [LIVE]

Detailed violation analysis across multiple dimensions:

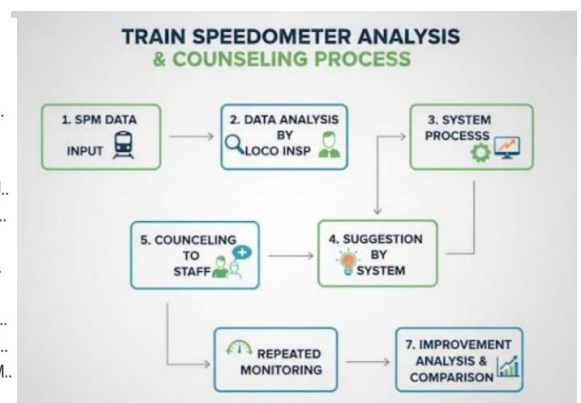
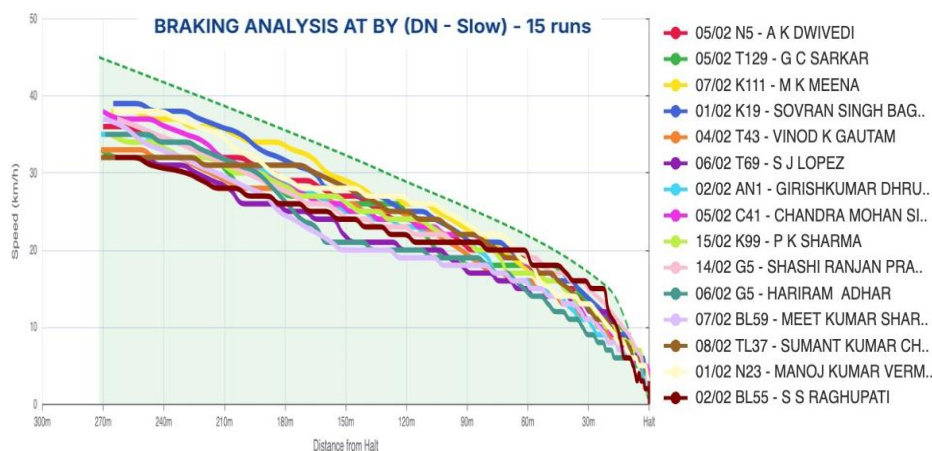
- Station-wise, section-wise, and time-period-wise breakdowns
- Violation type: >60 Kmph, >40 Kmph, and other defined thresholds
- Platform entry speeds for Suburban trains
- Identification of LPs without SPM analysis for 15+ days or 3+ months

Intelligent Exception Alerts [PLANNED]

When CRTMS detects speed violations, abnormal braking patterns, or sub-par counselling performance, nominated CLIs will receive push notifications for timely review — enabling proactive safety management before incidents occur.

TSR & Signal Mapping Integration [LIVE]

Integration of Temporary Speed Restrictions (TSR) and signal locations enables precise, location-based violation detection for improved safety oversight.



SIGNAL-BASED PRECISION MONITORING · PLANNED EXPANSION

A comprehensive digital signal database — covering signal lists, inter-signal distances, sighting distances, and GPS coordinates — will allow CRTMS to:

- ▶ Identify the exact signal where a train halted or a speed violation occurred
- ▶ Detect repeated abnormal braking locations across the section
- ▶ Link Suburban AWS operations for focused review and corrective action

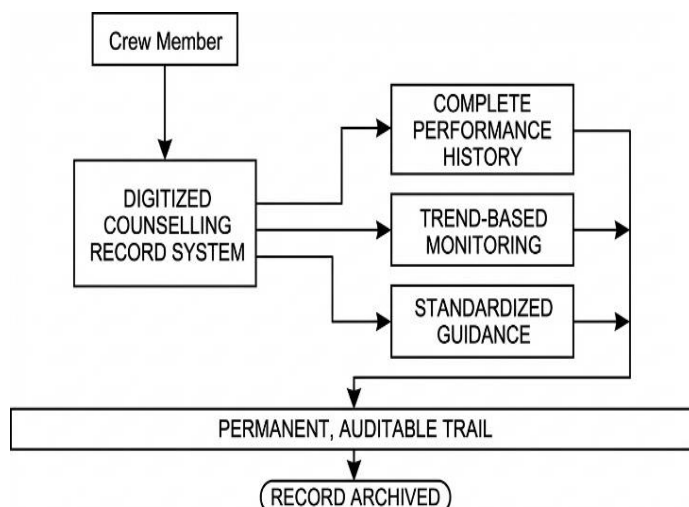
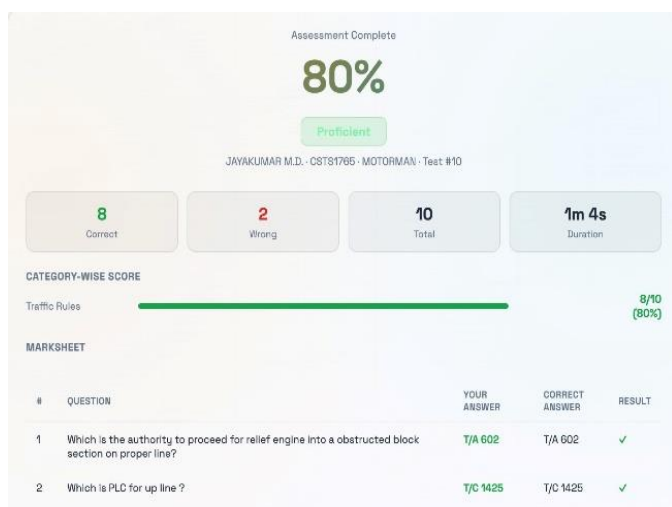
04 ▶ COMPUTER-BASED CREW COUNSELLING (CBC)

Digital Counselling Platform [LIVE]

A fully digitized counselling record system providing complete performance history, trend-based monitoring, and standardized guidance — creating a permanent, auditable trail for every crew member.

CBC Mobile App — Progressive Web App [PLANNED]

The planned PWA brings CRTMS counselling to the field: accessible on mobile devices, operable offline, with automatic synchronization — empowering CLIs to supervise effectively anywhere on the section.



INTEGRATED · INTELLIGENT · TRANSPARENT

Strengthening Safety · Enhancing Efficiency · Adding Accountability

“Mission Zero SPAD.”



Mumbai Division · Central Railway · www.crtms.in